A COMPUTATIONAL DESIGN SCIENCE APPROACH FOR INTER-VEHICLE NETWORK COMMUNICATION USING LEO SATELLITE INTERNET

Version 1.04/2025.08.13

Lappeenranta–Lahti University of Technology LUT

Degree Programme in Software Engineering, Master's thesis

2026

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Examiners: Title First name and Last name (optional degree)

Title First name and Last name (optional degree)

Abstract

Lappeenranta–Lahti University of Technology LUT

LUT School of Engineering Science

Degree Programme in Software Engineering

Aleksandar Ivanov

A computational design science approach for inter-vehicle network communication using LEO satellite internet

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2026

19 pages, 3 figures, 1 tables and XX appendices

Examiners: Title First name and Last name (optional degree) and Title First name and Last name (optional degree)

Keywords: key1, key2

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Acknowledgements (optional)

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The author, Aleksandar Ivanov, takes full responsibility for the content of this thesis and has reviewed and edited the content generated by the possible use of AI tools.

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1 Introduction

1.1 CBRN Defense Field

CBRN defense field encompasses the military and civil measures taken to protect against chemical, biological, radiological, and nuclear threats. The field includes a wide range of activities, such as detection, identification, decontamination, and medical countermeasures (Otřísal, 2014). CBRN events can pose a significant physical and psychological hazard to the public and pose environmental and economic impact (Ministry of Health Malaysia, 2025). Fast, coordinated and accurate response in the first critical hour or two is crucial to minimize the impact of such events (Joint Emergency Services Interoperability Programme (JESIP), 2016). Providing accurate information in a timely manner is essential for effective decision-making and response (Ministry of Health Malaysia, 2025).

Coordination between the various agencies and personnel involved in CBRN defense is essential. Numerous software solutions have been developed to facilitate communication and information sharing among CBRN defense teams (Marques, Duarte & Lobo, 2025; Environics Oy, 2026; Bruhn NewTech, 2021; Observis Oy, 2026). A common pattern among these solutions is the near real-time consolidation and visualization of data from multiple sources, such as sensors, first responders, command centers and various governmental databases. However, the effectiveness of these solutions is limited by the communication infrastructure available in the affected area. Additionally, any delays introduced by the communication infrastructure, protocols, and data processing can add up to significant delays in the overall response time. In the case of CBRN events, where every second counts, these delays can have serious consequences.

1.2 LEO Satellite Networks

One of the more recent developments in communication infrastructure is the commercial availability of low earth orbit (LEO) satellite internet. LEO satellite networks have the potential to provide high-speed, low-latency internet access to remote and underserved areas (Duan & Dinavahi, 2021). This technology has the potential to improve the communication abilities of CBRN defense teams by providing a communication channel in remote areas or areas with delibierately disrupted communication infrastructure (Abels, 2024).

One of the leading providers of LEO satellite internet is Starlink, which as of September 2024 has launched 6426 satellites into orbit and has 14 million active subscribers globally, making it the leader by a wide margin in both of these metrics. Starlink subscriptions are widely available across the globe in both personal and business plans (Koosha, Madani &

Ardakani, 2025). The relatively low subscription cost, easy of acquisition, low-latency, high bandwidth and coverage make it a compelling option as the communication infrastructure for CBRN defense teams.

1.3 QUIC Protocol

QUIC is a multiplexed, encrypted and low-latency transport protocol that is designed with the goal of improving transport performance and security of internet applications. QUIC is a cross-layer protocol and operates at the Transport, Security and Application layers of the OSI model (Langley et al., 2017).

1.4 Motivation and Context

This thesis aims to address these challenges by proposing a computational design science approach for inter-vehicle network communication using LEO satellite internet. The proposed approach leverages the advantages of LEO satellite internet, such as low latency and high bandwidth, to enhance the communication capabilities of CBRN defense vehicles. By improving the speed and accuracy of information sharing, the proposed approach has the potential to significantly enhance the effectiveness of CBRN defense operations.

1.5 Objectives and Scope of This Work

2 Theoretical background

Give a brief introduction to what each chapter is about. Write at least two sentences under the chapter heading. As a general guideline, the paragraph has to contain at least two sentences.

2.1 Low Earth Orbit (LEO) Satellite Networks

Low Earth Orbit (LEO) satellites are satellites that orbit the earth at a speed higher than that of earth's rotation. As a result of that higher speed, LEO satellites appear to continuously circulate around the earth. In order to provide global coverage a constellation of LEO satellites is required. (Sun, 2014)

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4 Conclusions

The conclusions explain how well your research achieved its objectives, what its findings were and what they mean in a wider perspective and for the future. The conclusions should examine how your findings differ from or coincide with those of previous studies. Analyse the impact of your research: its theoretical or practical contribution and wider societal importance. In addition, mention possible limitations of your study and research topics that should be dealt with in the future.

Remember that if or when someone other than your supervisor reads your thesis, they will most likely read the introduction and conclusions first. Below is an example of using multi-level lists in the LaTeX environment. Regarding lists, it is good to note that their use is not recommended in the conclusions, but that the conclusions are presented as a paragraph text.

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Appendix 1 Text processing and layout in a thesis

Good text processing skills make writing your final thesis and using this template easier. Therefore, you should ensure you have sufficient basic skills to edit long documents with LaTeX before you start. This involves applying LaTeX code structures, understanding automatic referencing and knowing how to divide your text into sections.

The essential thing is to understand the following basics of LaTeX:

- LaTeX needs a compiler. Compiler can be stand-alone in the computer or cloud-based, such as Overleaf
- LaTeX will remove all the unnecessary spaces and linebreaks. When you want to start a new paragraph, make one empty line into your code, LaTeX will start the paragraph.
- Do not enter any numbering manually. LaTeX has efficient automatized tools for this. Section, page, figure, and table numbering are automatically applied in this template. They keep the numbers in the right order even if you modify, add or remove content.
- Do not add hyphenation at the end of a line manually. LaTeX automatic hyphenation tool can be used in this template. It should work with both English and Finnish languages.

Line spacing, font, margins, alignment, page numbering and headings

Official layout guidelines state that the line spacing should be 1.5, except for the abstract and possible direct citations, where the spacing is 1. You can choose from two fonts: Times New Roman (12 pt) or Arial (11 pt). This template has been written in Adobe Times Roman 12 pt.

Leave the following margins:

- top and left 35 mm.
- bottom and right 20 mm.

The page number of the title page is 1, but the page numbering should not be visible before the first page of the table of contents. Place the page numbers at the top, either centred or in the right-hand corner. Page numbering ends on the final page of the reference list: appendices do not have page numbers unless the appendix is multiple pages. For a single-page appendix, use the command `\thispagestyle{empty}` to remove the page number.

Always use section styles in your thesis (`\section{}`, `\subsection{}`, `\subsubsection{}`).

Always place section headings (`\section{}`) on a new page. Apply `\clearpage` command before the section heading starts the section from the new page. If you add large figures or tables, remember to check that the empty space after the figure or image covers no more than 20% of the page.

The heading numbering starts from the introduction and continues consecutively. Do not place a period after the heading number. Do not number the heading of the reference list at the end of the thesis.

The thesis should include no more than three heading levels, and the headings should progress logically. If you need even more detailed subheadings, do not number them; leave them out of the table of contents. Concise headings that describe the text sufficiently are the best. You can use question marks or exclamation marks but do not add a period if the heading is a regular sentence.

A heading cannot follow a heading. Always write something between them. For instance, there must be text between headings 1 and 1.1 with two or more sentences.

Lists are a good way to express things clearly. Use the same type of bullet or symbol in lists throughout your thesis. A section should never end in a list. There should always be two or three sentences after a list.

Appendix 2 References

The text must include references to the sources you use. LUT University applies the Harvard referencing style, also called the author-date style, with in-text referencing and a detailed reference list at the end.

The purpose of a reference list is to provide sufficient information on a source used in the study, allowing the reader to consult the original source for further information. The reference enables the reader to find detailed information on the source easily in the list of references. You should refer to the original and most recent sources. If no new studies have been published on the topic in question, also older ones may be used.

Referring to a source means that you explain the contents of the source material in your own words. Direct citations, on the other hand, are placed inside quotes “Quote”. Plagiarism or using another person’s original material without appropriate referencing is not allowed.

Referencing technique

In the Harvard system, the citation is placed in parentheses directly in the text to indicate the passage that has been cited from another source. Place the citation before the period that ends the sentence when it refers only to the sentence in question (Geldart, 1985). If you are referring to more than that one sentence, introduce the source you are summarizing or paraphrasing at the beginning of the paragraph. Then, refer back to the source when needed to ensure your reader understands you are still using the same source. In-text references, specifying page numbers, are not used unless a specific part of the text is directly referred to or it is a direct quotation. Please note that the author does not always have to be an individual but can also be, for example, an organization. If the publication’s author does not appear in the source, the name of the publication is referred to instead of the author’s name.

Author’s name cited in the text (direct)

Direct citation is used when the author’s name(s) is a part of the sentence structure, with the year in parentheses. The following is obtained by using `\citet{key}` command.

> Allen & Tildesley (1987) claims that …

More than three authors for a work

> Chowdhury et al. (2017) in their work …

On some special occasions, all the authors (more than three) can be cited, but this form should be considered only at the *special* case by using `\citet*{key}` command

> Bizon, Shattuck, Swift, McCormick & Swinney (1998) showed ...

Author's name not cited in the text (indirect)

Indirect citation is used when the author's name(s) is not mentioned directly in the sentence structure. The indirect citation will appear at the relevant point in the sentence or at the end of the sentence in parentheses. The following is obtained by using `\citep{key}` command.

> Earlier research claims that ... (Savage, 1992).

Two authors with postnote

> At the theoretical approaches of ... (Allen & Tildesley, 1987, p. 102).

More than three authors for a work

> ..., (Veleva et al., 2015).

Combined citation with three entries using the command `\citep{key1,key2,key3}`

> According to previous studies, the presented conclusions are consistent (Miller, 1997; Dalton, Petri & Pontuale, 2005; Allen & Tildesley, 1987).

In many situations, it is justified to combine the content of several sources, and then it is recommended to use content-specific indirect references. For example, in the following way:

> "In literature we can find challenges relating to a wide variety of issues including environmental sustainability (Dagiliute & Liobikiene, 2015), business needs (Veleva et al., 2015), food production (Chowdhury et al., 2017), urban development (Joss, 2010; Swapan, 2016; Freeman, McMahon & Godfrey, 2017; Waite et al., 2017), and education (Zidansek, 2007; Biberhofer & Rammel, 2017)." (Talikka, 2018)

A wide variety of sources can be used in the text. In addition to the above, Master's theses (Hyvärinen, 2019), websites (NASA, 2015), interviews (Kottarainen, 2007, interview 6.6.2007), technical reports (Laine, Leppänen & Nurmi, 2009), booklets (Swetla, 2015), manuals (R Core Team, 2018), proceedings (Stepney & Verlan, 2018), standards (ISO 9004, 2000), patents (Virtamo & Aalto, 1996) or AI-language models (OpenAI, 2023) can be used.

All the bibliography information is given in the References. If you refer several works published by the same author in the same year, add lowercase letters (a, b, c. . .) after the publication year to distinguish the sources. Use the same alphabetical notation also in the list of references. There are several referencing and citation styles. It is essential to use the same style throughout the thesis consistently. Examples and detailed instructions on referencing:

- LUT Academic Library's instructions on how to cite electronic documents
- Aalto University citation guide
- Harvard referencing, University of Sheffield

Appendix 3 Tables, figures, equations, numbers, symbols, and abbreviations

It is a good idea to illustrate your text with figures and tables. Figures and tables must have captions and consecutive numbering. The captions of tables are placed above the table, and those of figures are below the figure. Refer to the figures and tables in the text body, preferably before you introduce them, and align them with the text body. In many theses, the alignment of figures and tables has been centered and without wrapping text around them, but the author can select the formatting options as long as they are coherent in the whole work. LaTeX tries to place all floating objects (figures and tables) at the top of the page or in the best place and to avoid multiple floating objects on the same page. Because of this, for several floating objects in series, their position can be shifted several pages forward. It is also possible for floating objects to move to the next section. Because of this, several consecutive floating objects should be avoided, and enough text should be added between the floating objects. In this template `\usepackage[section]{placeins}` setting is used to force floating objects to stay in the appearing section. You can also use `\FloatBarrier` command to set a position limit for floating objects.

Remember to add alt text (alternative text) to your figures and tables to ensure accessibility. Alt text is read by a designated reader and can be viewed even when the image cannot be displayed on the page. In LaTeX the alt text has to be set manually and you should make sure it describes the object sufficiently and understandably. You can add the alt text by using the command `\pdftooltip{Object}{Alt-text}`.

Tables

Give your table number and caption, and refer to them in the text. Place the caption above the table, name its columns, and mention the units applied, as in Table 1 below. Avoid empty columns or rows. The recommended font size is 10.

Table 1: Sensor measurements.

Voltage U [V]	Pressure p [Pa]
0.984	0
2.252	150
2.772	300
3.181	450
3.615	600
3.817	750
4.088	900

Figures, charts, graphic elements

Images help illustrate your text. The text should contain a reference to every figure. Number your figures and place a caption underneath – not inside the figure.

You should use a software program like Excel or Matlab to draw charts. Charts should be clear and easy to understand. Use a white background. A background grid is allowed if it does not make the figure difficult to interpret. Variables and measurement points should be clearly visible. Name the axes and their units. An example script to produce a Figure 1 with Matlab is presented in Appendix 6. To maintain full accessibility of the document, you should use embedded fonts in the vector figures. The example script will not produce a figure with embedded fonts, and the jpg or png formats would be better with respect to document accessibility. However, the figure quality is reduced.

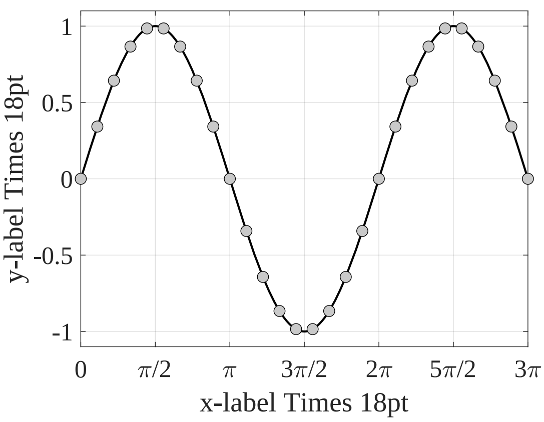

Figure 1: Illustration of Matlab example figure.

Create as many of the figures yourself as you can. Use the same font as in the text body and equations. Try to maintain the font size in the figure as close as possible to the main text font size. If you use images created by someone else, remember to cite them correctly. Remember that images are copyrighted works, the use of which must always be authorized by the author. Captions need to be in the same language as the text body.

Figure 2 shows a gas fermenter in a laboratory configuration. The image is positioned at the beginning of the page, and with the command `\FloatBarrier`, the following text is forced to appear after the image.

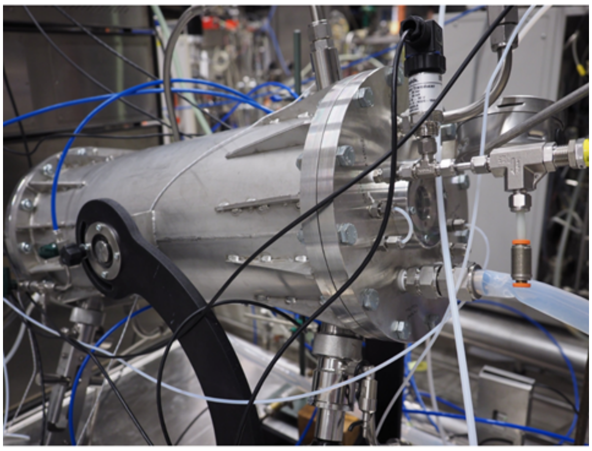

Figure 2: Gas fermentor (VTT 2020, LUT image bank).

Do not end a paragraph in a figure or table. Add text underneath, such as comments on the figure. Large figures, tables, long equations, and other supporting material can be appended.

Numbers, symbols and equations

Numbers in the text are usually approximations. Their accuracy depends on the observational error. Include only significant figures in the results. Interim results should include at least two figures more to avoid round-off errors. Present large and small figures in powers of ten 10^n, where n should preferably be divisible by three.

Equations and other mathematical expressions must consist of standardised symbols if one exists. You may use other symbols only if there are no applicable standardized or established ones. Variables and constants are denoted using *slanted style*, vectors are denoted using **bold regular style**, and abbreviations and dimensionless numbers are denoted using regular style. See the SI-unit style guide.

Explain the symbols in an equation when you use them for the first time. Write each equation clearly on its own line and indent it. Number your equations consecutively or by paragraphs so that the number is in parentheses on the right side of the equation and aligned to the right. You can refer to an equation only after you have presented it, with certain exceptions, such as if the object you are referring to is far ahead. As a simple example, the equation of state of an ideal gas can be used

$$pv = RT, \tag{1}$$

where p is pressure [Pa], v is specific volume [m^3/kg], R is the gas constant [J/(kgK)] and T is temperature [K].

The SI-unit style should be followed. Here are some common guidelines for using variables, constants, and units correctly:

- Write scalars in italics and vectors in bold, not in italics. Don't write dimensionless quantities in italics.
- Don't use $*$ multiplier in your equations. Present the variables to be multiplied in the equations in consecutive order, placing the numerical scalar value first, without any multiplier sign. Exceptions: for 1.452×10^6 or when you place numerical values into the equations in the example calculations, use `\times` command for a multiplier. In the latter case, remember to show the units as well.
- Use only single-character variables to avoid mistakes with previous point. You may use subscripts to separate variables.
- Write subscripts upright unless there is a need to italicise them (they are variables). Write abbreviated subscripts and numerals e.g., as follows: $\Delta\sigma_{\mathrm{w}}$, σ_1, or σ_{min}. For instance, in the summation $\sum_i^\infty x_i$, the subscript i must be italicised because it represents a variable.
- If you wish to express a change in, e.g. pressure Δp, write Δ in a regular font. In some cases, $\mathit{\Delta}$ may also be variable and should be italicised. $\mathrm{\pi} = 3.14159$ is the ratio of a circle's circumference to its diameter. π may be the pressure ratio.
- Do not italicise mathematical operators such as $\sin(x)$, $\cos(x)$, $\log(y)$, or $\ln(y)$.
- Distinguish absolute values as follows: "variable_=_value_unit", with the exception of a percentage sign after a numeral, e.g. a = 5.2 mm, γ = 97.7%.
- Use a decimal point "." in accordance with international standards. In contrast, a decimal comma is used in theses written in Finnish. This also applies to figures and tables.

List of symbols and abbreviations

List symbols and abbreviations and their definitions that are not common knowledge separately on their own page before the table of contents. Divide them into groups: Latin characters, Greek characters, subscripts, superscripts, abbreviations, and finally, dimensionless numbers. Give the page the heading Symbols if there are no abbreviations or Abbreviations if there are no symbols.

When you use a symbol or abbreviation in the text body for the first time, introduce it to the reader, for example, as follows: "The concept design for manufacturing and assembly (DFMA) is…". After this, you can use only the abbreviation, and the reader can verify its meaning from the abbreviation list. Do not add concepts to the list of symbols and abbreviations you do not mention in your text. Don't include standard, well-known elements or species (e.g., oxygen O_2 or carbon dioxide CO_2) definitions in the abbreviations.

Appendix 4 Appendices to the thesis

Appendices may include, e.g., interview questions, survey forms, or other content relevant to the work but not necessary to include in the text body. All appendices are public and should not contain any confidential information.

In your text body, refer to the appendices by adding their title in parentheses (Appendix 1) where relevant. Give all appendices a title based on their content and list them in the table of contents in the order in which they are referred to in the thesis.

Single-page appendices do not require page numbering. Multiple-page appendices do.

Appendix 5 Publishing the thesis

LUT's degree regulations state that Bachelor's and Master's theses are public documents. They are published in the LUTPub repository, and related instructions are available on the library website.

Together with the first examiner, make sure that the commissioner of your thesis is aware of the publicity requirements from the very beginning of the discussions. The student writing the thesis and the thesis commissioner are responsible for ensuring that there is no confidential material in the thesis. The background material of the thesis may contain confidential content. The background material is not public.

All theses in LUTPub must fulfill accessibility requirements. Your text must be as legible as possible to readers. Remember:

- to add PDF meta-data at the beginning of main.tex according to Figure 3 with your own information.
- to use LaTeX commands for headings,
- embed hyperlinks into your text or a description of the linked content; do not include URL addresses,
- add/check the alt text of your figures and tables describing briefly the main content in writing.
- to refer to each figure and table in the text, and verbally describe in the text the key content presented in the figures and tables.

```
%%%%%%%%%%%%%%%%%%%
%% Give first the meta-data for the PDF/A
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% IMPORTANT!!!!                                    %
%% Mandatory data, update with your own data        %
%% This data will appear in the pdf file properties %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
\begin{filecontents*}[overwrite]{\jobname.xmpdata}
\Title{Title of the work}
\Author{Author_firstname Author_lastname}
\Language{en-GB}
\Subject{The abstract or short description.}
\Keywords{template\sep thesis\sep LaTeX}
\end{filecontents*}
```

Figure 3: Example to define PDF meta-data in latex-system.

At the moment (09/2023), LaTeX text system and its add-ons (packages) can only produce pdf files according to accessibility criteria to a limited extent. Using the LaTeX system, it is not possible to produce a final product that meets all accessibility criteria and works universally.

Due to deficiencies, the automatic read-out of the PDF file may have problems, especially with images, tables, and equations. For this reason, it is especially important that the contents of floating objects (images, tables, and equations) are described as clearly as possible in the body text.

Appendix 6 Matlab script for example figure

```
%% This script will generate an example figure and it will export the
    figure in jpg, png and pdf format
% Jouni Ritvanen, 2023

close all
clear

fs=18;                  % Font size
fc=[0.79 0.79 0.79]; % marker face color (light grey)

x=0:pi/9:3*pi;     % creating x-vector

figure  % create figure
fplot(@sin,[0,3*pi],'k','LineWidth',1.5)     % plotting sin(x) function
hold on % this allows to add next plot into the figure
grid on % adding grid
plot(x,sin(x),'o',...            % plotting discrete sin(x) points
     'MarkerSize',8,...          % setting marker size to 8
     'MarkerEdgeColor','k',... % setting marker edge color to black
     MarkerFaceColor=fc)         % setting marker face color to fc'

set(gca,'fontname','times')    % setting font to Times
set(gca,'fontsize',fs)         % setting font size to fs
ylim([-1.1 1.1])               % setting y-axes limits
xlabel('x-label Times 18pt')   % setting x-axes label
ylabel('y-label Times 18pt')   % setting y-axes label
xticks(0:pi/2:3*pi)            % setting x-ticks every pi/2
% setting x-ticks labels
xticklabels({'0','\pi/2','\pi','3\pi/2','2\pi','5\pi/2','3\pi'})
%% this section will adjust the paper size to correspond the figure size
% i.e. the output will be the size of Matlab figure, not e.g. A4
pps=get(gcf,'paperposition');    % gets figure size and position
set(gcf,'papersize',pps(3:4));  % sets paper size to correspond figure
    size
set(gcf,'paperposition',[0 0 pps(3:4)]);     % sets figure to fill paper
%% this section will save the figure in jpeg and pdf formats
print -djpeg -r600 example_name % output will in resolution of 600
print -dpng -r600 example_name  % png resoluutiolla 600
print -dpdf example_name         % vector pdf output
```

Appendix 7 Bibliography entries in .bib file

```
%% Reference examples

% Article, two authors
@article{DAGILIUTE,
title          = {University contributions to environmental
    sustainability: challenges and opportunities from the Lithuanian case
    },
journal        = {Journal of Cleaner Production},
volume         = {108},
pages          = {891-899},
year           = {2015},
doi            = {10.1016/j.jclepro.2015.07.015},
author         = {Renata Dagiliute and Genovaite Liobikiene}}
% Article, five authors
@article{VELEVA,
title          = {Understanding and addressing business needs and
    sustainability challenges: lessons from Devens eco-industrial park},
journal        = {Journal of Cleaner Production},
volume         = {87},
pages          = {375-384},
year           = {2015},
doi            = {10.1016/j.jclepro.2014.09.014},
author         = {Vesela Veleva and Svetlana Todorova and Peter Lowitt
    and Neil Angus and Dona Neely}}
% Article, four authors
@article{CHOWDHURY,
title          = {Key sustainability challenges for the global
    phosphorus resource, their implications for global food security, and
    options for mitigation},
journal        = {Journal of Cleaner Production},
volume         = {140},
pages          = {945-963},
year           = {2017},
doi            = {10.1016/j.jclepro.2016.07.012},
author         = {Rubel Biswas Chowdhury and Graham A. Moore and Anthony
     J. Weatherley and Meenakshi Arora}}
% Article, one author
@article{swapan,
author         = {Abu Yousuf Swapan},
journal        = {Journal of Urban and environmental Engineering},
number         = {2},
pages          = {221--232},
title          = {Holistic strategy for urban design: {A} microclimatic
    concern to sustainability},
```

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volume           = {10},
doi              = {10.4090/juee.2016.v10n2.221232},
year             = {2016}}
% Article, three authors
@article{freeman,
author           = {Rachel Freeman and Chris McMahon and Patrick Godfrey},
title            = {An exploration of the potential for re-distributed
    manufacturing to contribute to a sustainable, resilient city},
journal          = {International Journal of Sustainable Engineering},
volume           = {10},
number           = {4-5},
pages            = {260-271},
year             = {2017},
doi              = {10.1080/19397038.2017.1318969}}
% Article, six authors
@article{WAITE,
title            = {Global trends in urban electricity demands for cooling
     and heating},
journal          = {Energy},
volume           = {127},
pages            = {786-802},
year             = {2017},
doi              = {10.1016/j.energy.2017.03.095},
author           = {Michael Waite and Elliot Cohen and Henri Torbey and
    Michael Piccirilli and Yu Tian and Vijay Modi}}
% Article, one author
@article{ZIDANSEK,
title            = {Sustainable development and happiness in nations},
journal          = {Energy},
volume           = {32},
number           = {6},
pages            = {891-897},
year             = {2007},
doi              = {10.1016/j.energy.2006.09.016},
author           = {Aleksander Zidansek}}
% Article, two authors
@article{Biberhofer,
title            = {Transdisciplinary learning and teaching as answers to
    urban sustainability challenges},
journal          = {International Journal of Sustainability in Higher
    Education},
volume           = {18},
number           = {1},
pages            = {63-83},
year             = {2017},
doi              = {10.1108/IJSHE-04-2015-0078},
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author          = {Petra Biberhofer and Christian Rammel}}
% Article, five authors
@ARTICLE{Biz98,
author          = {C. Bizon and M. D. Shattuck and J. B. Swift and W. D.
   McCormick and H. L. Swinney},
title           = {Patterns in 3{D} Vertically Oscillated Granular Layers
   : Simulation and Experiment},
journal         = {Physical Review Letters},
year            = {1998},
volume          = {80},
pages           = {57--60},
number          = {1},
doi             = {10.1103/PhysRevLett.80.57}}
% Book
@BOOK{All87,
title           = {Computer Simulation of Liquids},
publisher       = {Oxford University Press Inc.},
year            = {1987},
author          = {M. P. Allen and D. J. Tildesley},
address         = {New York},
ISBN            = {0-19-855645-4}}
% In book
@INBOOK{CitekeyInbook2,
author          = {D. Geldart},
title           = {Elutriation},
editor          = {J. F. Davidson and R. Clift and D. Harrison},
booktitle       = {Fluidization},
edition         = {2nd edn.},
year            = {1985},
publisher       = {Academic Press},
address         = {London},
ISBN            = {0-12-205552-7},
pages           = {383--412},
chapter         = {11}}
% On book
@INBOOK{Joss,
author          = {S. Joss},
title           = {Eco-cities: a global survey 2009},
editor          = {C.A. Brebbia and S. Hernandez and E. Tiezzi},
booktitle       = {The Sustainable City VI: Urban Regeneration and
   Sustainability},
year            = {2010},
publisher       = {WIT Press},
address         = {Southampton},
ISBN            = {978-1-84564-432-1},
pages           = {239--250},
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@INCOLLECTION{Sav92,
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title            = {Numerical simulation of Couette flow of granular
    materials; spatiotemporal coherence and 1/f noise},
booktitle        = {Physics of Granular Media},
publisher        = {Nova Science Publishers},
year             = {1992},
editor           = {D. Bideau and J. A. Dodds},
series           = {Les Houches Series}}
% In proceedings
@INPROCEEDINGS{Dal05,
author           = {F. Dalton and A. Petri and G. Pontuale},
title            = {Stress fluctuations and the solid/fluid transition in
    a sheared granular bed},
booktitle        = {Powders and Grains},
year             = {2005},
editor           = {R. García-Rojo and H.J. Herrmann and S. McNamara},
address          = {Stuttgart, Germany, Jun. 18-22, 2005},
pages            = {353--355}}
% In proceedings
@INPROCEEDINGS{Gol98,
author           = {I. Goldhirsch},
title            = {Kinetics and Dynamics of Rapid Granular Flows},
booktitle        = {Physics of Dry Granular Media},
year             = {1998},
editor           = {H. J. Hermann and J. P. Hovi and S. Luding},
volume           = {350},
series           = {NATO ASI Series E},
pages            = {371--400},
publisher        = {Kluver Academic Publishers}}
% PhD thesis
@PHDTHESIS{Mil97,
author           = {B. J. Miller},
title            = {Stress fluctuations in continuously sheared granular
    media},
school           = {Duke University},
address          = {Durham, NC},
year             = {1997}}
% DSc thesis
@PHDTHESIS{Talikka,
author           = {Marja Talikka},
title            = {Recognizing required changes to higher education
    engineering programs' information literacy education as a consequence
    of research problems becoming more complex},
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school          = {Lappeenranta University of Technology},
address         = {Lappeenranta, Finland},
type            = {Doctoral thesis},
year            = {2018},
URL             = {https://urn.fi/URN:ISBN:978-952-335-238-4}}
% Master's thesis
@THESIS{Hyv19,
author          = {Joonas Hyvärinen},
title           = {Techno-economic evaluation of carbon capture
    technologies integrated to flexible renewable energy system},
school          = {Lappeenranta-Lahti University of Technology LUT},
address         = {Lappeenranta, Finland},
type            = {Master's thesis},
year            = {2019},
URL             = {https://urn.fi/URN:NBN:fi-fe2019100731528}}
% Website
@MISC{CitekeyMisc,
title           = {Pluto: The 'Other' Red Planet},
author          = {NASA},
howpublished    = {[Website]},
URL             = {https://www.nasa.gov/nh/pluto-the-other-red-planet},
year            = 2015,
note            = {[Cited: 2023-03-06]}}
% Technical report
@TECHREPORT{Lai09,
author          = {J. Laine and T. Leppänen and O. Nurmi},
title           = {The influence of special effects on the perception of
    printed products},
institution     = {VTT Technical Research Center of Finland},
year            = {2009},
type            = {Research report},
series          = {VTT-R-04966-09},
address         = {Espoo}}
% Interview
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type            = {[Phone interview] 6.6.2007},
author          = {Brian Kottarainen},
year            = {2007},
note            = {Interviewer Outi Opiskelija, the recording is in the
    author's possession}}
%k Booklet
@booklet{CitekeyBooklet,
title           = {Canoe tours in {S}weden},
author          = {Maria Swetla},
howpublished    = {Distributed at the Stockholm Tourist Office},
year            = {2015}}
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% Manual
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title          = {{R}: A Language and Environment for Statistical
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author         = {{R Core Team}},
organization   = {R Foundation for Statistical Computing},
address        = {Vienna, Austria},
year           = {2018}}
% Proceedings
@proceedings{CitekeyProceedings,
editor         = {Susan Stepney and Sergey Verlan},
title          = {Proceedings of the 17th International Conference on
   Computation and Natural Computation, Fontainebleau, France},
series         = {Lecture Notes in Computer Science},
volume         = {10867},
publisher      = {Springer},
address        = {Cham, Switzerland},
year           = {2018}}
% Standard
@TECHREPORT{standardi,
author         = {{ISO 9004}},
title          = {Quality magement systems -- Guidelines for performance
    improvements},
institution    = {International Organization for Standardization},
year           = {2000},
type           = {Standard},
address        = {Geneve, Switzerland}}
% Patent
@MANUAL{US5503249,
type           = {Patent, US-5503249},
author         = {J. Virtamo and S. Aalto},
organization   = {Kone Elevator, SW Baar},
title          = {Procedure for controlling an elevator group},
year = {1996}}
% AI language model
@MISC{ChatGPT,
type           = {[Large language model]},
title          = {ChatGPT, Mar 14 version},
author         = {OpenAI},
URL            = {https://chat.openai.com/chat},
year           = {2023}}
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```
@article{Otrisal2014,
```

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  author  = {Pavel Ot\v{r}\'{\i}sal},
  title   = {The Concept of {CBRN} Defence and Possibilities of its
      Application in the {Czech Armed Forces}},
  journal = {Journal of Defense Resources Management},
  year    = {2014},
  volume  = {4},
  number  = {2},
  pages   = {1--4},
  url     = {https://www.longdom.org/open-access/the-concept-of-cbrn-
      defence-and-possibilities-of-its-application-in-theczech-armed-
      forces-21873.html}
}

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